INFT-1205 – Data Communications & Networking II

Capstone Project

Overview

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By: Group 1

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1. Summary

This document outlines the complete redesign Datacore Solutions' network infrastructure, following the expansion from one floor to two. The redesign required a new IPv4 address plan, VLAN segmentation, inter-VLAN routing, Layer 2 security, EtherChannel between floors, and redundant ISP connectivity using static and floating static routes.

Our team designed and deployed a secure and scalable network, ensuring communication between the departments, accommodating future growth, and aligning with the enterprise network standards.

2. Network Topology

The physical network is located at 1 King Street, Toronto, and consists of 2 floor switches, the Datacore Solutions edge router, the Primary Rogers Communications ISP router, and the Backup Bell Canada ISP router. The topology is as follows:

- **DatacoreRouter:** The core router of Datacore Solutions. It connects the ISP routers with dedicated WAN connections, to Datacore Solutions' internal LAN, provides DHCP services for the IT and HR departments, and allows inter-VLAN routing using a trunk port and Router-on-a-Stick.
- **RogersRouter:** The Primary ISP router from Rogers Communications. It connects to the DatacoreRouter on the 101.1.1.0/30 network.
- **BellRouter:** The Backup ISP router from Bell Canada. It connects to the DatacoreRouter on the 101.1.2.0/30 network.
- **FL1:** The floor 1 switch hosts the VLANs for the IT, Customer Support, and Sales and Marketing departments. It connects to the DatacoreRouter via a trunk port, and to FL2 via a 3-lane EtherChannel.
- **FL2:** The floor 2 switch hosts the VLANs for the HR, Management, System Admins, Accounting, and Legal departments. It connects to FL1 using the same 3-lane EtherChannel.

The physical topology is fully outlined in **Topology.vsdx**.

3. IPv4 Address Plan

VLSM of VLANs

We used Datacore Solutions designated private address block of **10.160.1.0/23**. This network provides 510 usable hosts. Using Variable Length Subnet Masking (VLSM), subnets were created for each department based on the host requirements.

Department	Floor	Default Gateway	Subnet Mask	Hosts
IT	1	10.160.0.1	255.255.255.128	100
Customer Support	1	10.160.0.129	255.255.255.192	45
Sales & Marketing	1	10.160.0.193	255.255.255.192	60
HR	2	10.160.1.1	255.255.255.192	45
Management	2	10.160.1.193	255.255.255.224	30
System Admins	2	10.160.1.65	255.255.255.192	60
Accounting	2	10.160.1.129	255.255.255.192	55
Legal	2	10.160.1.225	255.255.255.224	25

Subnets were grouped by floor to maintain logical structure and simplify troubleshooting. The largest subnets were allocated first to prevent fragmentation and ensure minimal IP waste.

WAN Links

The ISP connections use /30 subnets, which allows for 2 usable hosts:

- DatacoreRouter uses 101.1.1.1 (first host) and RogersRouter uses 101.1.1.2 (last host).
- DatacoreRouter uses 101.1.2.1 (first host) and BellRouter uses 101.1.2.2 (last host).

The complete addressing plan is documented in **Addressing.xlsx**.

4. Company Requirements

VLAN Segmentation & Inter-VLAN Routing

Each department was assigned its own VLAN, ensuring proper VLAN segmentation. Inter-VLAN routing was implemented on the DatacoreRouter using Router-on-a-Stick, allowing reliable communication between departments.

DHCPv4 Deployment

DHCP services were required for the IT (floor 1) and HR (floor 2) departments. DHCP pools were configured on the DatacoreRouter, providing automatic IPv4 information. The first 10 addresses were excluded, allowing for up to 10 static IP configurations, while still respecting the required hosts for the IT and HR departments.

EtherChannel Between Floors

A 3-lane EtherChannel link aggregation was configured between FL1 and FL2. This ensures that the bandwidth increases by a factor of 3, while also providing redundancy in case of individual link failure. The EtherChannel operates using LACP for stability and automatic negotiation.

ISP Redundancy

Datacore Solutions required a primary connection to Rogers Communications, and a backup connection to Bell Canada. We implemented:

- A static route for the primary Rogers path
- A floating static route with an administrative distance of 10 for the Bell path

This provides automatic failover in the primary ISP becomes unavailable.

Layer 2 Security

The FL1 and FL2 switches were configured with the following layer 2 security features:

- Port security violation set to restrict, preventing unauthorized users.
- DHCP snooping enabled, preventing rogue DHCP servers.
- Dynamic ARP Inspection (DAI) to block ARP spoofing.
- BPDU Guard to protect STP.
- Secure Passwords on FL1, FL2, and DatacoreRouter.
- SSH on DatacoreRouter.
- All unused ports places in a Parking Lot, and shutdown.
- All access ports set as untrusted, and all trunk ports set as trusted ports.

These measures protect the network from internal threats, misconfigurations, and unauthorized access.

5. Design Decisions

VLSM Allocation

We designed the addressing plan by allocating the largest subnets first, preventing fragmentation and ensuring minimal IP waste. Subnets were grouped by floor to maintain the logical structure and simplify troubleshooting.

Router-on-a-Stick

We implemented Router-on-a-Stick for inter-VLAN routing, as it is an efficient, cost-effective and scalable method, allowing up to 50 VLANs for inter-VLAN communication. This approach centralizes routing on the DatacoreRouter, keeping switch configurations much simpler.

Layer 2 Security Approach

Layer 2 security was implemented uniformly across both switches. This approach provides consistent protection across all departments, and reduces the risk of internal attacks, such as MAC flooding, ARP spoofing, or unauthorized device connections.

6. Conclusion

The redesigned Datacore Solutions network is secure, scalable, and aligned with the company's requirements. By implementing VLAN segmentation, inter-VLAN routing, DHCP services, EtherChannel, redundant ISP connectivity, and Layer 2 security, our team was able to deliver a robust solution.

The final implementation ensures:

- Communication between all departments.
- Redundancy through the primary and backup ISPs, as well as EtherChannel configuration.
- Security against internal threats, misconfigurations, and unauthorized access.
- Efficient utilization of the 10.160.1.0/23, through VLSM.

The network is designed to be maintainable and scalable allowing Datacore Solutions infrastructure to continue growing.